SwinPA-Net: Swin Transformer-Based Multiscale Feature Pyramid Aggregation Network for Medical Image Segmentation

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***Abstract*— The precise segmentation of medical images is one of the key challenges in pathology research and clinical practice. However, many medical image segmentation tasks have problems such as large differences between different types of lesions and similar shapes as well as colors between lesions and surrounding tissues, which seriously affects the improvement of segmentation accuracy. In this article, a novel method called Swin Pyramid Aggregation network (SwinPA-Net) is proposed by combining two designed modules with Swin Transformer to learn more powerful and robust features. The two modules, named dense multiplicative connection (DMC) module and local pyramid attention (LPA) module, are proposed to aggregate the multiscale context information of medical images. The DMC module cascades the multiscale semantic feature information through dense multiplicative feature fusion, which minimizes the interference of shallow background noise to improve the feature expression and solves the problem of excessive variation in lesion size and type. Moreover, the LPA module guides the network to focus on the region of interest by merging the global attention and the local attention, which helps to solve similar problems. The proposed network is evaluated on two public benchmark datasets for polyp segmentation task and skin lesion segmentation task as well as a clinical private dataset for laparoscopic image segmentation task. Compared with existing state-of-the-art (SOTA) methods, the SwinPA-Net achieves the most advanced performance and can outperform the second-best method on the mean Dice score by 1.68%, 0.8%, and 1.2% on the three tasks, respectively.**

***Index Terms*— Dense multiplicative connection (DMC) module, local pyramid attention (LPA) module, medical image segmentation.**

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I. Introduction

DEEP-LEARNING-BASED image segmentation plays a critical role in medical image analysis and has achieved impressiveimpressive performance in several medical tasks [1], [2], [3], such as the segmentation of organs and tissues in various types of microscopic images [4], [5], polyp segmentation in colonoscopy images [6], [7], and skin lesion segmentation in dermoscopy images [8]. Accurate and efficient medical image segmentation can be employed not only to identify the features of the lesion areas but also provide a reliable basis to aid clinical diagnosis and condition research [9], [10], [11], [12] and has potential application value in surgical navigation and disease progression monitoring.

In recent years, convolutional neural networks (CNNs) such as ResNet [13] have shown powerful feature extraction capabilities. Subsequently, the fully convolutional network (FCN) [14] creatively replaced the fully connected layer in CNN with the convolutional layer to preserve the spatial information of the image for semantic segmentation tasks [15]. Inspired by FCN, a typical U-shaped network, named U-Net [16], is proposed to overcome problems on medical image segmentation. By using the encoder–decoder structure and skip connections, the network can learn the multiscale features properly and has performed well in many medical image segmentation tasks, quickly becoming a baseline for comparing the state-of-the-art (SOTA) medical image segmentation algorithms. With the popularity of U-Net, some innovative models based on the U-shaped architecture have been proposed and achieved good results, such as UNet++ [17], SegNet [18], DenseUNet [19], KiU-Net [20], and CAB U-Net [21]. However, the convolutional-based deep learning networks have inherent defects, such as the difficulty of accurately defining the boundary features and the lack of global context information, which limits its further improvement in the performance of biomedical image segmentation tasks.

To address the above issues, several novel approaches have been developed in recent years, such as the Mumford–Shah model, contrastive learning, and pyramid structure. Methods based on the Mumford–Shah model [22], [23] extract accurate boundary information of the original image through piecewise smoothing functions and optimization strategies, while methods based on contrastive learning [24], [25] regulate the distance between semantic information by exploring the

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pixel-level relationship between features. However, the time complexity of these methods is often very high, and it is difficult to accurately segment different types of lesions with large differences. The pyramid structure [26], [27], [28], [29] obtains multiscale contextual information by utilizing different convolution rates or multiresolution input feature maps. For example, Deeplab [26] adopts multiple convolution rates with different receptive fields to improve the multiscale information extraction ability of the model. SSPP [27] utilizes different stride pooling operations to generate multiresolution feature maps. However, it is difficult for these methods to effectively identify small local features, and the perceptual field of these methods is not well adjusted to obtain the best features for lesions of different sizes at each stage.

Inspired by the huge achievements of transformers in natural language processing (NLP) tasks [30], several deep learning networks with transformer backbone have been utilized in the computer vision (CV) domain [31]. The vision transformer (ViT) [32] is the first transformer architecture designed for image classification, which encodes the input image as two-dimensional flattened patches with the location embeddings, producing convincing results on ImageNet by pretraining on a large dataset. Following the technical route of ViT, data-efficient image transformer (DeiT) [33] can learn the local features through the distillation token, and it can achieve good performance in smaller datasets because of the data-efficient training strategy. Although the above transformer-based architecture can reach SOTA in some image recognition tasks, these networks generally have high demands on computer hardware and face the lack of generalization abilities. To solve these problems, Swin Transformer [34] proposes a self-attention mechanism based on shifted windows, which improves the generalization of the model and greatly reduces the computational complexity through cross-window connections. It has achieved SOTA performance in different kinds of tasks, such as image classification, object detection, and semantic segmentation.

In this article, motivated by the success of Swin Transformer and other novel methods mentioned earlier, we adapted this work to the area of medical image segmentation and introduced two designed modules into the U-shaped network with Swin Transformer as backbone to solve the problems that still exist in medical image segmentation. For instance, there is a large gap between different types of lesions in various medical image datasets, with lesions having similar shapes and colors to the surrounding tissues and low contrast, as shown in Fig. 1. Specifically, the two modules, named dense multiplicative connection (DMC) module and local pyramid attention (LPA) module, are proposed to aggregate the multiscale context information of different kinds of medical images. The DMC module cascades the multiscale semantic feature information through multiplicative feature fusion, guiding the network to focus on feature information of different scales, which can reduce the interference of shallow feature noise and improve the expression of features. Furthermore, the DMC module can help the network to better identify lesion edges in different sizes and solve the problem of size changes. The LPA module uses a pyramid module to aggregate attention cues. In each layer, the feature map is split into different sizes and dimensions, and then, the local as well as global attention maps are obtained and integrated. Through the integration of attention maps with different dimensions, the network can be guided to focus on the target area, which is conducive to enhance the distinction of semantic features and reducing the impact of low contrast.

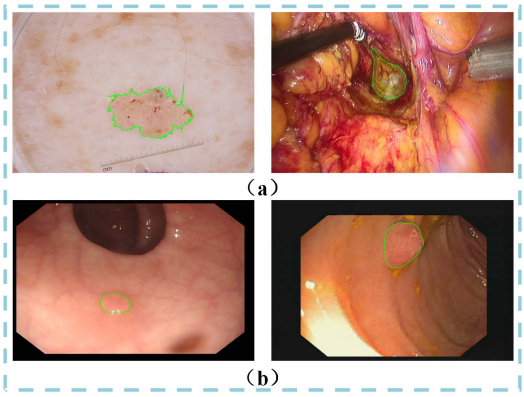


Fig. 1. Difficulties in medical image segmentation. (a) Cases where the lesions between different medical image datasets are quite large. (b) Cases where the pixel variations between the lesion and surrounding tissues are small. The areas enclosed by the green line indicate the lesion part.

With the above descriptions, we name the method as Swin Pyramid Aggregation network (SwinPA-Net). The proposed SwinPA-Net is tested on three challenging medical image segmentation tasks: polyp segmentation in colonoscopy images, skin lesion segmentation in dermoscopy images, and the segmentation of loose connective tissue in laparoscopic images. The first and second tasks are based on public benchmark datasets, while the third task is based on a clinical private dataset.

The contribution of this work can be summarized in the following three areas.

1) A U-shaped network named SwinPA-Net is proposed, which introduces two designed modules into the network with Swin Transformer as backbone. During encoding, the proposed model employs Swin Transformer to extract features, enabling the network to have global and dynamic receptive fields through self-attention and shifted window mechanisms. Also, the DMC and LPA module are proposed to enhance the feature extraction capability of the SwinPA-Net. The DMC module adopts a DMC method, which can reduce the interference of noise on the shallow features, so that the shallow information containing more boundary details can be obtained. The LPA module aggregates attention cues in different scales as well as guide the network to enhance semantic features and emphasize the target area, resulting in accurately locating the lesion. Moreover, the SwinPA-Net utilizes CNN as the decoder to restore and integrate the local and global feature maps to obtain more powerful and robust features.
2) By evaluating the proposed method on three challenging medical image segmentation tasks (polyp segmentation in colonoscopy images, skin lesion segmentation in dermoscopy images, and the segmentation of loose

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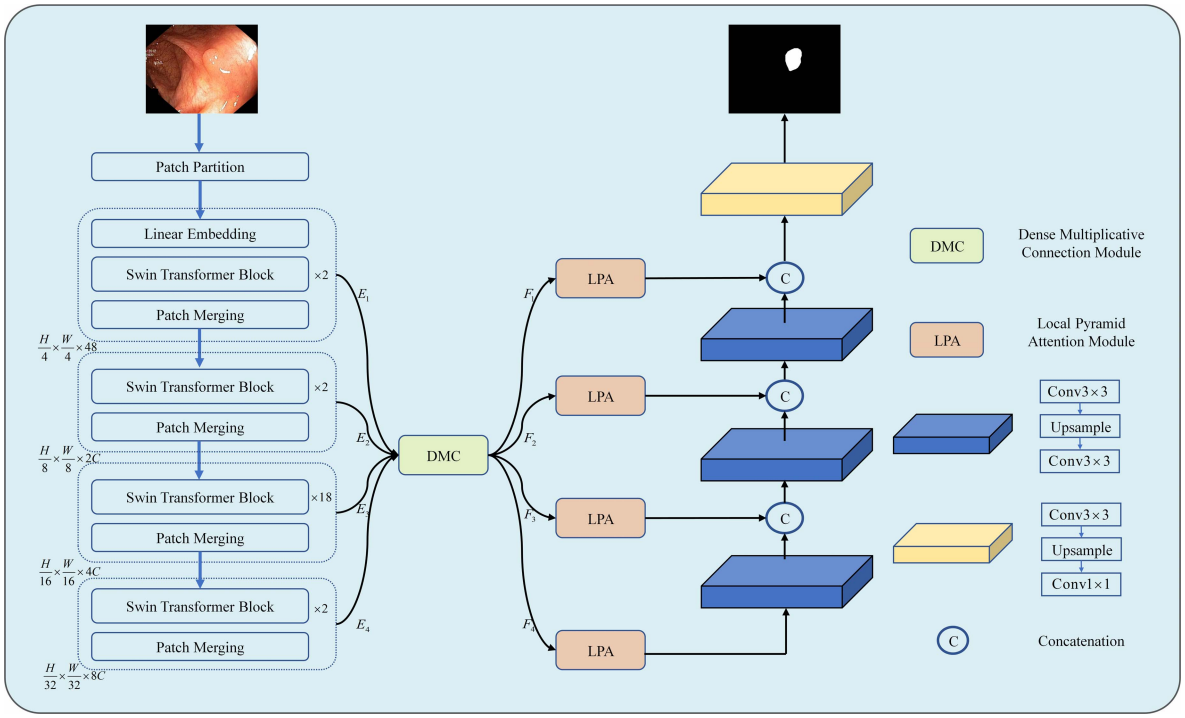


Fig. 2. Framework of the proposed SwinPA-Net. The medical image is fed into the encoder composed of Swin Transformer to obtain feature maps of different scales, and then, the multiscale feature information is merged through the proposed DMC module. Under the action of the LPA module, the attention mechanism is used to obtain the enhanced feature map of the target area. Furthermore, the feature map is upsampled by the decoder to the same resolution of the original image, and the final mask prediction is obtained. H and W denote the image height and width.

connective tissue in laparoscopic images), we demonstrate its good robustness and wide practicability. By introducing the proposed method, the mean Dice score was increased by 1.68%, 0.8%, and 1.2% compared with the second-best method on the three tasks, respectively. The experimental results show that SwinPA-Net achieves robust results and is competitive with other SOTA methods.

3) The Laparoscope Interoperative Vision (LIVis) dataset is created for recognizing loose connective tissue of the mesorectum to avoid damage to important tissues during total mesorectal excision (TME) surgeries. It contains 15 h of laparoscopic video and 1180 high-resolution images with 1920 × 1080 pixels, which are collected from the surgical robot at a hospital in Changsha, Hunan, China. The reference of these images contains pixel fractions for loose connective tissue of the mesorectum at different stages of the surgery, which are annotated and cross-verified by surgical experts. It is notable that the LIVis dataset can be used not only as a benchmark to evaluate the performance of the proposed method but also to provide reference for subsequent research.

The rest of this article is organized as follows. Section II describes the proposed image segmentation algorithm. The segmentation experiments on medical datasets are presented in Section III. Finally, we draw our conclusion in Section IV.

II. Method

A. Overview

In this section, the overall structure of the proposed SwinPA-Net is introduced in detail, as shown in Fig. 2. It is a U-shaped structure based on the decoder and encoder. First, we introduce the Swin Transformer structure used in the model and then introduce other important implementation details of the model, such as how the DMC module effectively integrates contextual multiscale feature information and how the LPA module combines local and global features.

B. Swin Transformer Encoder

A standard transformer structure first splits the input image equally into a series of nonoverlapping patches, and then, the pixels of these patches are linearly flattened into several 1-D vectors. Afterward, these vectors are passed through the multihead self-attention (MSA) module to calculate the self-attention and would be finally processed using Gaussian error linear unit (GELU) nonlinearity and multilayer perception (MLP) layers, while each module is preceded by a LayerNorm (LN) layer followed by a residual connection.

Due to the division of the image into small patches, the standard transformer needs to calculate relationships between these patches, where implies quadratically growing computational cost. Therefore, Swin Transformer employs window-based MSA (W-MSA) and shifted window-based MSA (SW-MSA), which changes the regular window partitioning method and uses a more efficient shifted window partitioning strategy, as shown in Fig. 3(a). Not only does it maintain efficient computation of nonoverlapping windows, but it also establishes connections between different windows. Two successive Swin Transformer blocks use W-MSA and SW-MSA modules, respectively, to establish the connection between different windows, as shown in Fig. 3(b). $\hat{z}^l$ and z^l represent the outputs of W-MSA and MLP in the lth layer, which can

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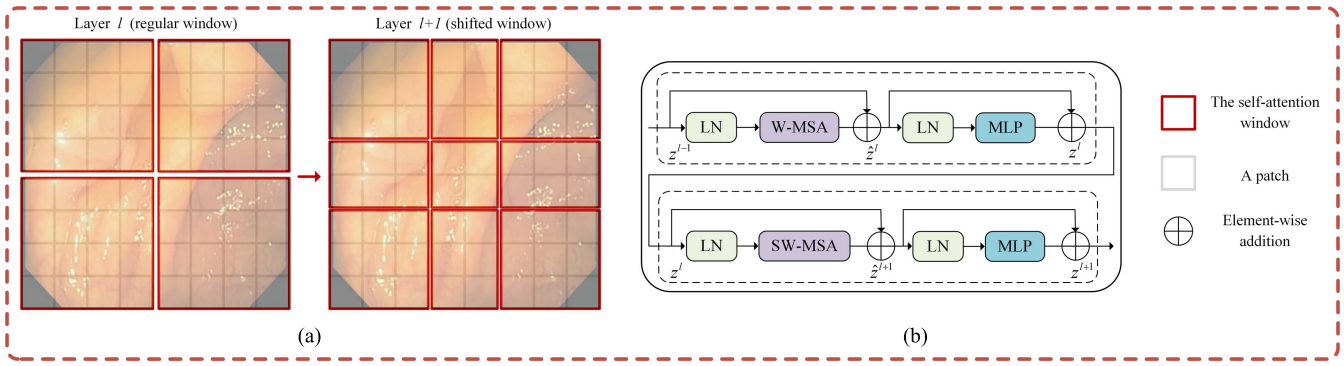


Fig. 3. Structure of Swin Transformer. (a) Windows for self-attention computations in adjacent layers l and $l+1$ are divided into regular window and shifted window. (b) Details of the Swin Transformer.

be expressed by the following equations:

$$\hat{z}^{l} = \text{W-MSA}(\text{LN}(z^{l-1})) + z^{l-1} \tag{1}$$

$$z^{l} = \text{MLP}(\text{LN}(\hat{z}^{l})) + \hat{z}^{l}. \tag{2}$$

By using the shifted window partitioning strategy, the outputs of the SW-MSA and MLP module can be expressed as

$$\hat{z}^{l+1} = \text{SW-MSA}(\text{LN}(z^{l})) + z^{l} \tag{3}$$

$$z^{l+1} = \text{MLP}(\text{LN}(\hat{z}^{l+1})) + \hat{z}^{l+1}. \tag{4}$$

In this article, considering the complexity and time efficiency, we adopt Swin-B as the encoder for feature extraction. The input image $X \in \mathbb{R}^{H\times W\times 3}$ is split into $(H/s)\times(W/s)$ nonoverlapping patches, where s represents the patch size, and these patches are flattened and projected to dimension c through a linear embedding layer. Next, these patch tokens are fed into the Swin Transformer and sequentially go through four stages to extract features. Each stage has several consecutive Swin Transformer blocks, including W-MSA and SW-MSA, and then uses the patch-merging module to downsample and double the dimensions of the features. The output of each stage is $(H/s)\times(W/s)\times C$, $(H/2s)\times(W/2s)\times 2C$, $(H/4s)\times(W/4s)\times 4C$, and $(H/8s)\times(W/8s)\times 8C$, respectively.

C. DMC Module

In the medical image segmentation tasks, we hope that the network can focus on the target area, but due to the different sizes of the targets, especially when some lesion areas or anatomical landmarks are relatively small, it is difficult to locate the target area accurately. The main reason is that the feature maps generated by the encoder would become increasingly coarse due to continuous downsampling. For a small target area, the boundary information in the deep feature maps would be turned into a few pixels or even lost altogether, which has a great impact on the accuracy of the medical image segmentation. A simple method is to input a large-size image to obtain a larger feature map, preserving the information of the lesion area as much as possible, but this would incur a huge computational cost. Another method is to fuse shallow features and deep features, where the shallow feature map contains a lot of detailed boundary information. However, since the shallow feature map simultaneously retains some noise and most feature fusion methods would mix some irrelevant information

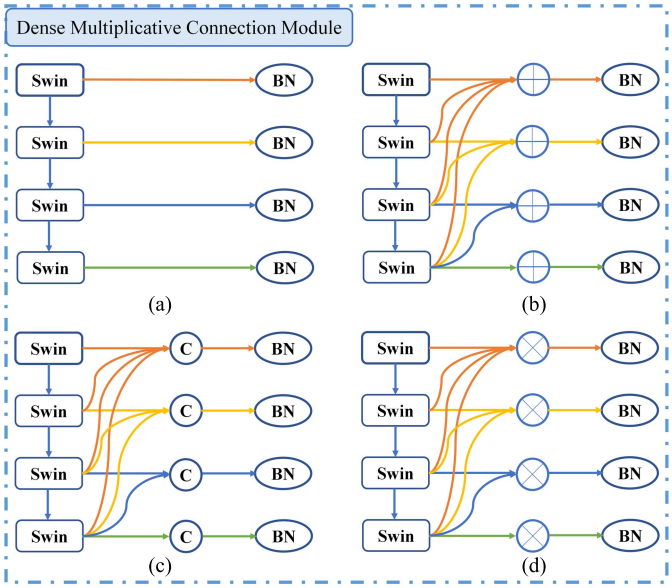


Fig. 4. Different styles of feature fusion. (a) Simple skip connection is used, but the features of different scales are not fused. (b) Dense addition connection (DAC) module: the features of different scales adopt the dense composition method of addition. (c) Dense concatenation connection (DCC) module: the features of different scales adopt the dense concatenation method of addition. (d) DMC module: the dense composition method of multiplication is adopted between the features. The composite connections are lines representing some simple operations, i.e., upsampling and 1×1 convolution.

when fusing shallow feature maps [35], [36], [37], the fused image generally contains a certain amount of noise, which also affects the segmentation results. Therefore, we propose the DMC module to reduce the noise of the shallow features, as shown in Fig. 4(d).

In the DMC module, we fuse feature maps of different scales with higher level semantic features to guide the network to focus on shallow and deep features to obtain a more comprehensive feature perception. At the same time, inspired by [38], the multiplication method can amplify the difference between the noise area and the normal area and reduce the noise interference caused by shallow features in the feature fusion process. Specifically, the output of the encoder is defined as $E_m (m = 1, \ldots, 4)$, and the output F_m of the feature fusion stage m is the product of the encoder output of this stage and the encoder output of the remaining $4 - m$ stage, which

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can be expressed as follows:

$$F_m = \prod_{i=m}^{4} g_i(E_i), \quad m \geq 1 \tag{5}$$

where $g(\cdot)$ represents the feature transformation, which uses 1×1 convolution to reduce the number of channels and upsampling to restore the resolution. Fig. 4 illustrates the difference between the multiplicative feature fusion method and three other fusion methods. Fig. 4(a) adopts the jump connection method of U-net without feature fusion, which would introduce the noise of shallow information, so that the effect of this method on image segmentation is not much improved. Fig. 4(b) and (c) utilizes the fusion method of addition and concatenation, respectively, so that shallow and deep features are merely combined, but using these two strategies would ignore the correlation between multiscale features. The above three ways can be expressed as follows:

$$F_2^{\text{mul}} = g_2(E_2) \cdot g_3(E_3) \cdot g_4(E_4) \tag{6}$$

$$F_2^{\text{add}} = g_2(E_2) + g_3(E_3) + g_4(E_4) \tag{7}$$

$$F_2^{\text{concat}} = \text{concat}(g_2(E_2), g_3(E_3), g_4(E_4)). \tag{8}$$

When the neural network backpropagates to calculate the gradient, the calculations can be expressed by the following equations:

$$\frac{\partial F_2^{\text{mul}}}{\partial g_2(E_2)} = g_3(E_3) \cdot g_4(E_4) \tag{9}$$

$$\frac{\partial F_2^{\text{add}}}{\partial g_2(E_2)} = 1 \tag{10}$$

$$\frac{\partial F_2^{\text{concat}}}{\partial g_2(E_2)} = \text{concat}(1, 0, 0). \tag{11}$$

It follows from (9) and (10) that when using the addition and concatenation methods, the gradient of each branch is constant and has no correlation with other branches. Therefore, the result of one branch does not affect other branches, and the network cannot smoothly learn the correlations between multiple branches. In the multiplication method, the gradient of each branch is not fixed and will be affected by other branches. In that case, if one of the branches cannot obtain excellent features, the use of multiplicative fusion will magnify the error of the branch, resulting in a large gradient. Therefore, the multiplication feature fusion method can greatly constrain each branch in the training process, so that each branch can obtain better features and different branches can promote each other, thereby filtering out the noise and obtaining more accurate prediction results.

D. LPA Module

In the medical image segmentation tasks, the desirable segmentation network would only focus on the target area of the lesion. However, due to the large difference between the types of lesions in various medical images and the small difference between the color of the lesion and the surrounding tissues, it is difficult for the network to accurately identify the location of the lesion. At the same time, the network needs to learn the complex structure of lesions with different sizes. In order to overcome these difficulties, attention mechanism is introduced to solve this problem, which can help the network locate interesting area as well as suppress useless information and has been widely used in deep learning [39]. For example, the bilateral attention decoder [40] adopts a lightweight bilateral decoder to capture semantic and spatial information more accurately. SAB Net [41] puts forward a semantic attention boosting (SAB) network that iteratively fuses contextual information across layers adaptively via semantic regularization. In this article, we propose the LPA module to guide the network to enhance the distinction of semantic features and emphasize the target area. The structure of LPA module is shown in Fig. 5.

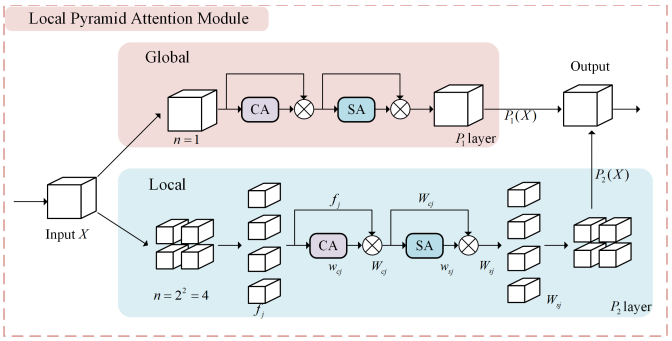


Fig. 5. Architecture of the LPA module. First, the feature map is divided into multiple feature vectors and the weight vector of each feature vector is obtained through the attention mechanism. Subsequently, the weight vector for the current dimension level is constructed by merging each weight vector and is applied to the original input feature map to obtain the attention map by elementwise multiplication. At last, the output is composed of the attention maps with different levels. The attention mechanism consists of two attention blocks: the CA and SA blocks.

The attention mechanism consists of two attention blocks: the spatial attention (SA) and channel attention (CA) block.

Not all regions in the image are equally important in contributing to the task, and thus, the SA module [39] is utilized as the SA block to find the task-relevant parts of the features by using interspatial information. Given the input feature, the SA A_S is defined as

$$A_S = \sigma(\xi(\text{Concat}(P_{\text{avg}}(X), P_{\text{max}}))) \tag{12}$$

where σ is the sigmoid function, and P_{max} and P_{avg} denote adaptive maximum pooling and adaptive average pooling across the channel functions, respectively. $\xi(\cdot)$ is a convolution with 7×7 filter.

The CA module measures the importance of features by assigning different weights to channels. In this article, we adopt "squeeze and excitation" block (SE layer) [42] as the CA module, which consists of a global average pooling layer and two fully connected layers, and the CA A_C is defined as

$$A_C = \sigma(\text{FC}_2\text{ReLu}(\text{FC}_1\text{Pool}_{\text{avg}}(X))) \tag{13}$$

where FC_1 and FC_2 represent two fully connected layers, and Pool_{avg} represents the average pooling layer.

Assume that an input feature map $X \in \mathbb{R}^{H \times W \times C}$ has height H, width W, and several input channels C, and then, the pyramid structure P_i is employed to learn the multiscale

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information of the feature map X, where P_i represents the ith layer pyramid. First, the global attention map is obtained through the first layer of the pyramid. Subsequently, the remaining layer of the pyramid splits the feature map into smaller dimensions and gets more accurate lesion boundary predictions through the local attention mechanism. Then, the local information is merged into the attention map of the current layer. Finally, the attention maps of different layers $P_i(X)$ are fused with each other to obtain the final attention map.

The detailed operation of P_i mainly consists of three stages. First, the input feature map X is split into n feature vectors of equal size along the H, W directions. The number of vectors n is based on the number of layers i and grows quadratically. As the number of pyramid layers i increases, the number of $n = i^2$ more refined feature vectors $f_j(j = 1, \ldots, n)$ would be gained. Second, after the feature vectors are obtained, the channel weight vectors $w_{cj}(j = 1, \ldots, n)$ can be obtained by learning each feature vector with the CA block and then multiplying the feature vector f_j with the CA vector w_{cj} to obtain the CA map $W_{cj}(j = 1, \ldots, n)$. Next, the CA map W_{cj} is sequentially passed through the SA block to obtain the corresponding SA matrix $w_{sj}(j = 1, \ldots, n)$, and then, the CA map W_{cj} is multiplied by the SA matrix w_{sj} to obtain the SA map $W_{sj}(j = 1, \ldots, n)$. Finally, based on the spatial location encoding information for the feature map at the time of splitting, all the SA maps W_{sj} are reintegrated with the inverse operation of splitting to obtain the attention map $P_i(X)$ corresponding to the current ith pyramid level, where the size of the attention map $P_i(X)$ is the same as that of the original feature map X.

Through the same operation, the network can be guided to find the attention maps $P_i(X)$ of different layers and would finally be merged to obtain the final attention map. This can effectively enhance the discriminative ability of each feature in feature learning and improve the ability of the network to recognize the target area. At the same time, we embed the proposed LPA structure into the four outputs of the DMC module.

E. Decoder

The decoder part combines the convolution blocks, upsampling, and skip connections to recover the resolution of the feature map stage by stage, as shown in Fig. 2. Specifically, at each stage, the intermediate feature map is first fused with the feature information cascaded from the same stage through the DMC and LPA modules output, adjusted by 3×3 convolution and upsampled using bilinear interpolation, and finally, the output of that stage is obtained using 3×3 convolution to improve the upsampling results. In particular, in the final stage of the encoder path, the feature map, which is processed from the previous stage, is recovered to the same size as the input image by the upsampling operation and 1×1 convolution adjustment of the number of channels.

F. Loss Function

Medical image segmentation usually has the problem of unbalanced foreground and background distribution and different sizes of lesion shapes. In order to better train and optimize our model, we combine Dice loss and binary cross entropy (BCE) loss to form a new loss function, which can be defined as

$$L_{\text{DICE}} = 1 - 2\frac{\sum_i y_i p_i + \varepsilon}{\sum_i y_i + \sum_i p_i + \varepsilon} \tag{14}$$

$$L_{\text{BCE}} = -\sum_i \left(y_i \ln(p_i) + (1 - y_i)\ln(1 - p_i)\right) \tag{15}$$

$$L = L_{\text{DICE}} + L_{\text{BCE}} \tag{16}$$

where y and p represent the real label of the image and the predicted result, respectively, and ε is a parameter set to 1 to improve stability. Through the combined loss function, the network can converge quickly and stably and can achieve good results on different medical image datasets. At the same time, in the comparison experiment, we used this loss function for training in different segmentation tasks for all comparison methods.

III. Experiments

The proposed SwinPA-Net and other SOTA segmentation methods were trained and tested under the same conditions on three challenging medical image datasets called the Polyp dataset, the ISIC 2018 dataset, and the LIVis dataset. Common metrics were used to evaluate the performance of the proposed segmentation method. Also, an ablation study was conducted to demonstrate the effectiveness of different modules in the SwinPA-Net.

A. Datasets and Implementation

1) Polyp Dataset: The polyp dataset consists of five public datasets named Kvasir [43], CVC-ClinicDB [44], CVC-ColonDB [45], EndoScene [46], and ETIS [47]. These datasets contain 2248 images of different sizes and are from different medical institutions. To ensure fairness, we adopted the division method in PraNet [6], where the images from Kvasir and CVC-ClinicDB datasets are randomly split into 90% for training (1450 images in total) and 10% for testing (162 images in total). To confirm the generalization ability of the proposed method, the CVC-ColonDB, EndoScene, and ETIS datasets (636 images in total) which did not appear in the training set were also used as the testing set.

2) ISIC 2018 Dataset: The ISIC 2018 dataset comes from the Lesion Boundary Segmentation task in ISIC-2018 challenge [48], which contains 2594 images from different types of skin lesions collected by several leading international clinical centers. These skin lesion images of different resolutions have corresponding manual labels as the reference standards for segmentation evaluation, and the labels have been continuously confirmed and fixed by experts. Fivefold cross-validation dataset was performed in the experiment, which split the dataset into 2075 images for training and 519 images for testing.

3) LIVis Dataset: LIVis dataset is a clinical private dataset collected from six live TME surgeries conducted on six patients using the surgical robot, as shown in Fig. 6. The TME surgery segregates the apical and visceral planes of the

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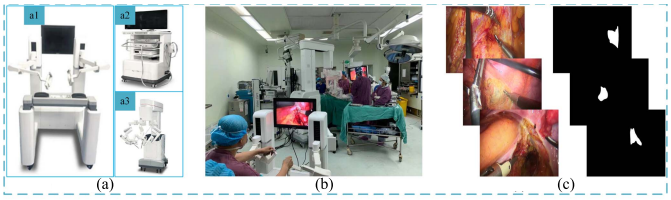


Fig. 6. Pipeline of data acquisition by the surgical robot. (a) Surgical robot is composed of the doctor console, camera system cart, and slave operator, as shown in (a1)–(a3), respectively. (b) During the TME operation, the doctor works on the surgical robot. (c) LIVis dataset is obtained after (b).

endopelvic fascia by dissociating loose connective tissue in the mesorectum, which minimizes the risk of a positive circumferential margin and has a strong prognostic significance for local recurrence [49], [50]. However, in order to avoid damage to important tissues such as blood vessels and nerves in the mesorectum when freeing loose connective tissue, it is very important to accurately segment the loose connective tissue in the field of vision during laparoscopy. Therefore, we created a dataset of loose connective tissue in different stages of surgery. Specifically, we sampled 1180 laparoscopic video images in stages from about 15 h of TME surgery videos, and the labels of these images have been carefully cross-verified by surgical experts, which can be utilized as the reference standards for the evaluation. In the experiment, the dataset was split into 844 images for training and 236 images for testing.

4) Implementation Details: Since the images in the above three datasets differ significantly in image size, resolution, and lesion characteristics, the proposed SwinPA-Net and other SOTA segmentation methods were trained and tested on different datasets in the experiments, respectively. Due to the limitation of GPU resources, instead of using the original image as the input of the network, the input images were resized to 384×384 pixels for both training and testing. Data augmentation was used to expand the training sets and enhance segmentation performance, including random flips and random shifts along different dimensions.

During the training stage, the AdamW optimizer [51] was adopted with a weight decay of 0.0005 and a maximum training epoch of 100, which was reduced by a factor of ten after 20 and 60 epochs, respectively, while the initial learning rate was equal to 0.0001. The network architecture was implemented in using the PyTorch framework, and all the experiments were performed on a standard workstation with a NVIDIA GeForce GTX 3090 GPU, an Intel Core 4215R CPU, and 128G RAM.

5) Performance Measures: Common evaluation metrics were adopted to quantify the segmentation performance of our proposed algorithm. For the Polyp dataset, we used the same evaluation metrics as in PraNet, including mean Dice coefficient (mDice), mean intersection over union (mIoU), and mean absolute error (MAE). Additionally, metrics such as mDice, mIoU, and recall were employed for quantitative evaluation on the ISIC 2018 dataset. For the LIVis dataset, we used the same evaluation metrics as CPFNet. The detailed definition of the metrics is as follows:

$$\text{mDice} = \frac{2 \times \text{TP}}{\text{FN} + \text{FP} + 2 \times \text{TP}} \tag{17}$$

$$\text{mIoU} = \frac{\text{TP}}{\text{FP} + \text{FN} + \text{TP}} \tag{18}$$

$$\text{Recall} = \frac{\text{TP}}{\text{FN} + \text{TP}} \tag{19}$$

where true positive (TP) represents the number of pixels that are correctly segmented as lesions, false positive (FP) represents the number of background pixels segmented as lesions pixels, true negative (TN) represents the number of pixels that are correctly segmented as normal tissue, and false negative (FN) is the number of true lesions pixels that are not segmented as such. MAE represents the average value of the absolute error between the true value and the predicted value. The smaller the value, the better the segmentation. For other indicators, larger values represent better segmentation performance.

B. Experiments on Polyp Segmentation

In this section, the proposed method is compared with other SOTA methods, including U-Net [16], U-Net++ [17], PraNet [6], CPFNet [52], Transfuse-S [53], and SANet [7] on the five polyp segmentation datasets. Following the settings of PraNet, the experiments with Kvasir and ClinicDB datasets were first conducted because they appeared in both the training set and the testing set. The results are shown in Table I, which suggest that the proposed SwinPA-Net achieves the best performance on these two datasets. Compared with the existing methods, our method improves the mDice evaluation metric when testing on the ClinicDB and Kvasir datasets, which are 4.2% and 2.7% higher than PraNet and 2.3% and 0.7% higher than Transfuse-S (a typical transformer-based method), respectively.

The trained networks were used to test the invisible datasets (ColonDB, EndoScene, and ETIS), and the results are shown in Table II. In particular, on the ETIS dataset with small polyp targets, our method improves by 1.2% and 13.4% compared to the SOTA methods (SANet and PraNet). By comparing the reference and segmentation results, we found that the SANet and Transfuse-S failed to segment small polyps or the segmented polyps were too large on some images, and they were not able to identify the difference between polyps and normal areas well, which is the reason for their poorer results. Additionally, for the large-area polyp target datasets (ColonDB and ETIS), the proposed method can also achieve excellent segmentation results, where the mean Dice score on the ColonDB dataset is 3.4% higher compared with the second-best method. On the EndoScene dataset, Transfuse achieves the best Dice coefficient. We believe in this experiment that the lesion areas of some images in the invisible polyp dataset are similar and have a certain continuity, as a result of which Transfuse may have a certain advantage due to the translation invariance of CNN. Furthermore, compared with the other two datasets, the number of samples in the EndoScene dataset is small (only 60 images), which means that a prediction error on one image can lead to huge fluctuations in the overall results. Therefore, the uncertainty of the results on this dataset and the properties of Transfuse itself lead to a slight advantage of Transfuse in the current results. Fig. 7 shows the examples

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TABLE I
QUANTITATIVE RESULTS OF THE TEST DATASETS KVASIR-SEG AND CLINICDB. BOLD INDICATE THE BEST RESULTS

Method	Kvasir			ClinicDB		
	mDice	mIoU	MAE	mDice	mIoU	MAE
U-Net [16]	0.818	0.746	0.055	0.823	0.755	0.019
U-Net++ [17]	0.821	0.743	0.048	0.794	0.729	0.022
PraNet [6]	0.898	0.840	0.030	0.899	0.849	0.009
CPF-Net [52]	0.907	0.856	0.028	0.910	0.856	0.011
Transfuse-S [53]	0.918	0.868	/	0.918	0.868	/
SANet[7]	0.904	0.847	0.028	0.916	0.859	0.012
Ours	**0.925**	**0.876**	**0.020**	**0.941**	**0.894**	**0.006**

TABLE II
QUANTITATIVE RESULTS OF THE INVISIBLE DATASETS COLONDB, ETIS, AND ENDOSCENE. BOLD INDICATES THE BEST RESULTS

Method	ColonDB			ETIS			EndoScene		
	mDice	mIoU	MAE	mDice	mIoU	MAE	mDice	mIoU	MAE
U-Net [16]	0.512	0.444	0.061	0.398	0.335	0.036	0.710	0.627	0.022
U-Net++ [17]	0.483	0.410	0.064	0.401	0.344	0.035	0.707	0.624	0.018
PraNet [6]	0.712	0.640	0.043	0.628	0.567	0.031	0.871	0.797	0.010
CPF-Net [52]	0.743	0.651	0.045	0.689	0.609	0.067	0.869	0.792	0.015
Transfuse-S [53]	0.773	0.696	/	0.733	0.659	/	**0.902**	**0.833**	/
SANet [7]	0.753	0.670	0.043	0.750	0.654	**0.015**	0.888	0.815	0.008
Ours	**0.807**	**0.726**	**0.029**	**0.762**	**0.684**	0.020	0.893	0.823	**0.007**

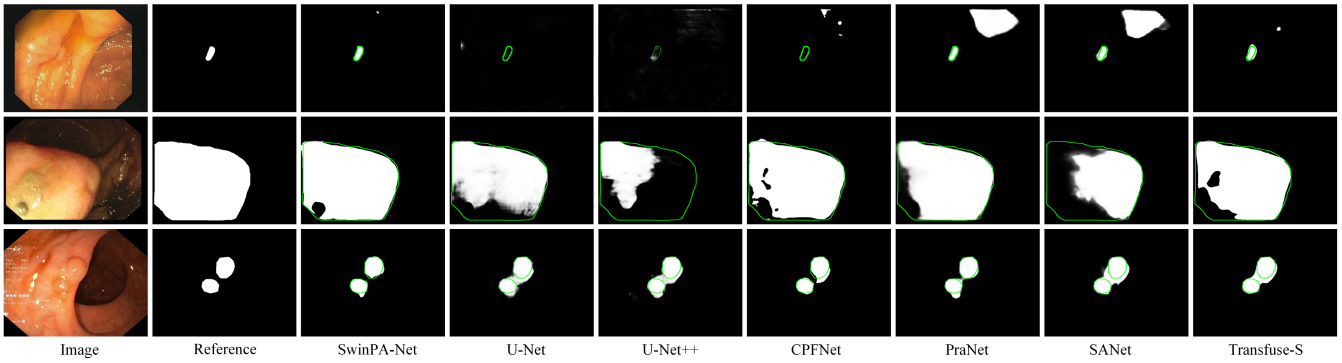


Fig. 7. Qualitative results of SwinPA-Net and other methods for the polyp segmentation task. The proposed method can improve the quality of segmentations considering its advantages in learning and generalization ability.

of segmentation results of the different methods on the different datasets, which indicate that the proposed method can achieve better edge predictions both in large-area polyps and small-area polyps datasets.

C. Experiments on Skin Lesion Segmentation

In this section, we evaluated the proposed method with other six excellent networks, such as U-Net [16], U-Net++ [17], PraNet [6], CPF-Net [52], Transfuse-S [53], and SANet [7] on the ISIC 2018 dataset. As shown in Table III, the proposed method achieves better segmentation with higher quality compared with CPF-Net and Transfuse-S. The proposed SwinPA-Net produces 0.910 on mDice, 0.854 on mIOU, and Recall of 0.946 with an improvement of 0.80%, 1.70%, and 3.40% compared with SOTA methods, respectively, and our proposed model achieves the best performances in all metrics. After analyzing the qualitative results, it is apparent that the method used can correctly predict the location and boundary of skin lesions and recognize the lesions from normal skins, as shown in Fig. 8.

D. Experiment on the LIVis Segmentation

In this section, we evaluate the proposed method on the LIVis dataset with six other excellent networks, such as U-Net [16], U-Net++ [17], PraNet [6], CPF-Net [52], Transfuse-S [53], and SANet [7]. As can be seen from Table IV, U-Net achieves the worst results, which may be due to the fact that U-Net does not make full use of the multiscale

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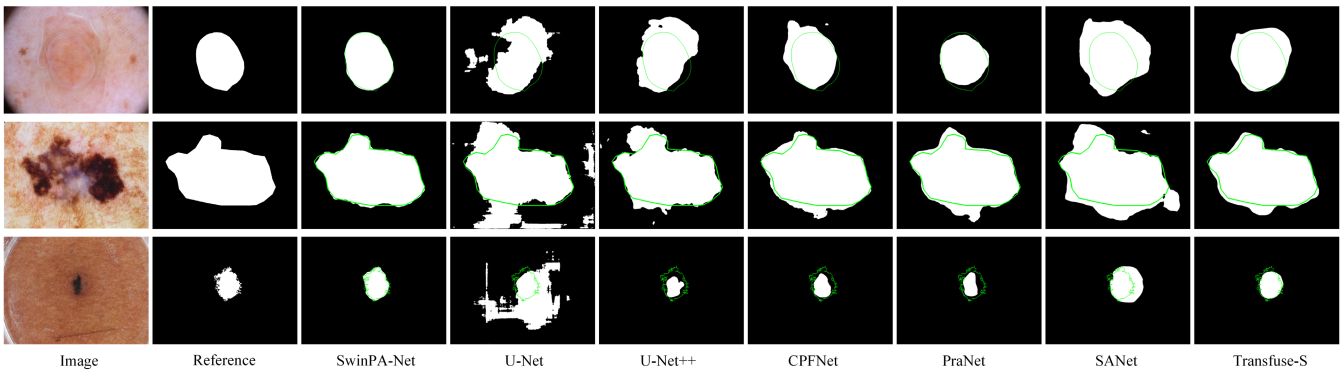


Fig. 8. Qualitative results of SwinPA-Net and other methods for the ISIC segmentation task. The proposed model can improve the quality of segmentations considering its advantages in learning and generalization ability.

TABLE III
QUANTITATIVE RESULTS OF THE ISIC 2018 DATASET. BOLD INDICATES THE BEST RESULTS

Method	mDice	mIoU	Recall
U-Net [16]	0.867±0.059	0.793±0.056	0.877±0.012
U-Net++ [17]	0.877±0.017	0.811±0.009	0.885±0.008
PraNet [6]	0.903±0.004	0.837±0.003	0.909±0.014
CPF-Net [52]	0.901±0.004	0.836±0.007	0.909±0.017
Transfuse-S [53]	0.902±0.001	0.835±0.005	0.912±0.007
SANet [7]	0.891±0.006	0.822±0.008	0.902±0.011
Ours	**0.911±0.003**	**0.854±0.004**	**0.946±0.010**

TABLE IV
QUANTITATIVE RESULTS OF THE LIVIS DATASET. BOLD INDICATES THE BEST RESULTS

Method	mDice	mIoU	Recall
U-Net [16]	0.865±0.013	0.776±0.018	0.859±0.009
U-Net++ [17]	0.888±0.007	0.811±0.008	0.895±0.004
PraNet [6]	0.871±0.004	0.781±0.005	0.881±0.003
CPF-Net [52]	0.874±0.003	0.787±0.004	0.865±0.003
Transfuse-S [53]	0.904±0.004	0.845±0.006	0.914±0.004
SANet [7]	0.832±0.017	0.723±0.013	0.846±0.015
Ours	**0.916±0.005**	**0.858±0.003**	**0.917±0.006**

features. PraNet and CPF-Net achieve approximately the same results, while the result of SANet is worse, which may be due to the smaller lesion area of the image. It is clear that our proposed SwinPA-Net achieves the best results on mDice, mIoU, and Recall with 91.6%, 85.8%, and 91.7%, respectively. Fig. 9 shows some examples of the segmentation results of different methods on the LIVis dataset.

E. Ablation Study

In this section, to prove the feature extraction capability and the effectiveness of different modules for the proposed SwinPA-Net (Model 4), an ablation study is conducted on the Kvasir dataset, as shown in Table V. The mIoU and mDice scores are employed to evaluate the segmentation performance in the ablation study.

Model 1 and 2 are built to compare the performance of traditional CNN and transformer structure to extract features. Model 1 changes the encoder of the U-Net structure to ResNet-152 as the backbone of feature extraction, and Model 2 uses Swin Transformer to extract features while maintaining the same structure as Model 1 in the decoder. By comparing with Model 1, we can see that the accuracy of segmentation has improved in Model 2, which proves the effectiveness of Swin Transformer as encoder to extract features.

Moreover, Models 3 and 4 are established to evaluate the usefulness of the DMC module and the LPA module, where the DMC module is added in Model 3 based on Model 2, and Model 4 adds the LPA module to Model 3. By comparing Models 2–4, we see that the introduction of the DMC module and the LPA module can considerably improve the mIoU and mDice scores. This is because the DMC module can fuse the features of different scales, while the LPA module can extract higher level information from local and global features, so that the model can better focus on useful features while suppressing less important features.

In addition, Models 5 and 6 are constructed to compare the effects of different feature fusion methods on the segmentation results, where Model 5 utilizes the additive fusion and Model 6 applies the concatenate fusion to replace the DMC module. By comparing Models 4–6, it can be seen the DMC module (Model 4) can obtain better feature expression through mutual constraints between multiscale features, which shows that the DMC module is more suitable for our method.

In order to verify the effectiveness of the LPA module with respect to the optimal number of layers and different parameter settings, we constructed Models 7–9, which are all identical to Model 4 except for the structure of the attention pyramid. Specifically, Model 4 adopts a two-layer LPA module, whereas Models 7 and 8 use one- and three-layer LPA modules, respectively. However, unlike the LPA module we have proposed, in the second layer of Model 9, the feature map is not split into smaller feature vectors for learning. By comparing Models 4, 7, and 8, we see that using a two-layer LPA module is better than a one-layer global attention, and when the number of pyramid layers is increased, the segmentation result is not better. The results show that the learning ability of one-layer

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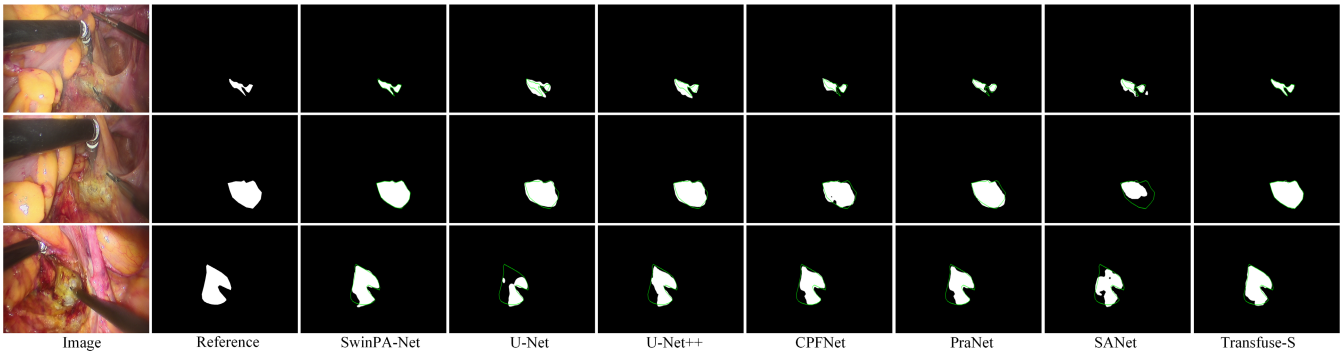


Fig. 9. Qualitative results of SwinPA-Net and other methods for LIVis dataset segmentation. The proposed model can improve the quality of segmentations considering its advantages in learning and generalization ability.

TABLE V
ABLATION STUDY OF THE PROPOSED METHOD

MODEL	Ablation Type									Results	
	CNN	Transformer	DMC	LPA	DAC	DCC	One-layer LPA	Three-layer LPA	Two same layers LPA	mDice	mIoU
Model 1	√									0.829	0.623
Model 2		√								0.895	0.845
Model 3		√	√							0.902	0.833
Model 4 (Proposed)		√	√	√						**0.925**	**0.876**
Model 5		√		√	√					0.914	0.857
Model 6		√		√		√				0.916	0.859
Model 7		√	√				√			0.920	0.869
Model 8		√	√					√		0.918	0.868
Model 9		√	√						√	0.914	0.856

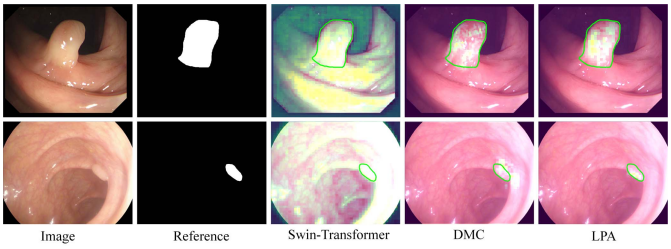


Fig. 10. Feature maps of DMC and LPA. The first to fifth columns of the figure are the original image, the reference label, the output feature maps of Swin Transformer, LPA, and DMC module, respectively.

LPA module is not good enough to learn the local features well. When the number of layers is deepened, it would be difficult for the model to learn higher dimensional information due to the limited training samples and image size, resulting in the deficient model performance. Compared with Model 9, Model 4 can achieve better results, which shows that it is effective to split the feature map into smaller feature vectors for learning, and the proposed LPA module does not obtain better performance by stacking parameters.

At the same time, the visualization results of DMC and LPA modules are shown in Fig. 10. The first to fifth columns of the figure are the original image, the reference label, and the output feature maps of Swin Transformer, LPA module, and DMC module, respectively. We can observe that although the output feature map of Swin Transformer can detect the information of the target area (the area outlines with green lines in the figure), it also contains a lot of noise. After utilizing the DMC module, the model can smoothly reduce the interference of noise in the shallow area as well as improve the ability of feature expression and can also pay attention to lesion areas of different sizes. In addition, the fifth column of Fig. 10 demonstrates that the LPA module captures the lesion area very well, which effectively solves the problem that the normal area and the lesion area are similar and improves the accuracy of segmentation.

IV. CONCLUSION

This article presents SwinPA-Net, which is a novel Swin Transformer-based multiscale feature pyramid aggregation network for medical image segmentation. The core idea of SwinPA-Net is to aggregate the context and boundary information of medical images through the Swin Transformer encoder and the designed DMC and LPA modules, while filtering out the noise and irrelevant information in the shallow features. The designed modules can effectively alleviate the problem that the lesions are difficult to locate accurately due to the large difference between the types of lesions in various medical images and the small gap between the color of the lesion and the surrounding tissues.

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The results of our extensive comparative experiments show that the proposed method can achieve SOTA results in most of the tested datasets, which fully demonstrates the excellent learning ability and strong robustness of SwinPA-Net in medical image segmentation. For example, in the polyp segmentation experiment, our method can segment the target area excellently in three unseen datasets, illustrating the strong adaptability of our method for different lesions. However, the method in this article has many parameters and requires a certain number of high-precision medical images to be annotated. In the future, our work would focus more on intraoperative segmentation, which implies higher requirements on the real-time performance. In addition, it is challenging to obtain plenty of accurately labeled medical images. Therefore, in the next step, we will try to explore a lightweight unsupervised segmentation method based on this method to meet the needs of intraoperative scenarios.

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